

Programas Lineales

Dado digrafo D , capacidades u , vértices r y s , encontrar x índices por $E(D)$ que

$$\begin{aligned} &\text{maximize } x^+(s) \\ &\text{sueto a } \begin{cases} x^+(v) = 0 & \forall v \notin \{r, s\} \\ x_e \leq u_e & \forall e \in E \\ x_e \geq 0 & \forall e \in E \end{cases} \end{aligned}$$

$$\begin{aligned} &\text{max } \textcircled{x} \\ &Ax \leq b \\ &x \geq 0 \end{aligned} \quad \text{--- (1)}$$


Obs Todo flujo rs -viable satisface (1) y toda sol viable de (1) es un flujo rs -viable.
Existen x en sol óptima si y solo si es flujo máximo.

Entérminar de matrices,

$$\begin{array}{c}
 \begin{array}{|c|} \hline V \setminus \{r, s\} \\ \hline \end{array} \\
 \begin{array}{c} \leftarrow x \rightarrow \\ \mu \\ \left[\begin{array}{cccccc} 0 & & & & & \\ \oplus & \ominus & \ominus & \oplus & \ominus & \\ & & \oplus & & & \end{array} \right] \\ \nu \end{array} \\
 \begin{array}{c} \left[\begin{array}{cccccc} 1 & & & & & \\ & & & & & 1 \end{array} \right] \\ \left[\begin{array}{cccccc} -1 & 0 & 1 & 0 & 1 & 0 \end{array} \right] \\ \leftarrow c \rightarrow \end{array} \\
 \begin{array}{c} \left[\begin{array}{c} 1 \\ \vdots \\ 0 \end{array} \right] \\ \mu \end{array} \\
 \begin{array}{c} \left[\begin{array}{c} 1 \\ \vdots \\ 0 \end{array} \right] \\ \mu \end{array}
 \end{array}$$

=

$$\begin{array}{c}
 \begin{array}{|c|} \hline 0 \\ \hline \end{array} \\
 \begin{array}{|c|} \hline 0 \\ \hline \end{array} \\
 \vdots \\
 \begin{array}{|c|} \hline 0 \\ \hline \end{array} \\
 \begin{array}{|c|} \hline \mu \\ \hline \end{array}
 \end{array}$$

<



max cx

$$\begin{array}{l}
 \text{s.t.} \left\{ \begin{array}{l} Ax = 0 \\ Ix \leq \mu \\ x \geq 0 \end{array} \right.
 \end{array}$$

Entender max cx
~~A~~ x ≤ b
 x ≥ 0

$$\max cx$$

$$Ax = 0$$

$$Ix \leq b$$

$$x \geq 0$$

$$\begin{aligned} Ax &\leq 0 \\ -Ax &\leq 0 \\ Ix &\leq b \\ x &\geq 0 \end{aligned}$$

$$\begin{aligned} \bar{A}x &\leq \bar{b} \\ x &\geq 0 \end{aligned}$$

$$\begin{bmatrix} A \\ -A \\ I \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ b \end{bmatrix}$$

$$\max cx$$

$$Ax \leq b$$

$$x \geq 0$$

(Primal)

$$\min yb$$

$$yA \geq c$$

$$y \geq 0$$

(Dual)

Duality
dual : $Cx \leq (yA)x = y(Ax) \leq yb$

DF : $Cx^* = y^*b$

Exm

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 - x_3 \\ \text{s.t.} \quad & x_1 - x_2 \geq 2 \\ & x_1 + x_3 = 5 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} \begin{bmatrix} -1 & 1 & 0 \\ 1 & 0 & 1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$$

μ_1
 μ_2

$$\begin{bmatrix} -2 \\ 5 \\ -5 \end{bmatrix}$$

$$\max \quad 3x_1 + 2x_2 - x_3$$

$$\begin{aligned} \text{s.t.} \quad & -x_1 + x_2 \leq -2 \\ & x_1 + x_3 \leq 5 \\ & -x_1 - x_3 \leq -5 \end{aligned}$$

$$x_1, x_2, x_3 \geq 0$$

$$\min \quad -2\mu_1 + 5\mu_2 - 5\mu_3$$

$$-y_1 + y_2 - y_3 \geq 3$$

$$y_1 \geq 2$$

$$y_2 - y_3 \geq -1$$

$$y_1, y_2, y_3 \geq 0$$

$$\begin{aligned} \max \quad & cx \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \end{aligned} \quad \left. \begin{aligned} \min \quad & y^T b \\ \text{s.t.} \quad & y^T A \geq c \\ & y \geq 0 \end{aligned} \right\}$$

$$\mu_1 = -y_1$$

$$\mu_2 = y_2 - y_3$$

$$\begin{aligned} \min \quad & 2\mu_1 + 5\mu_2 \\ \text{s.t.} \quad & \mu_1 + \mu_2 \geq 3 \\ & -\mu_1 \geq 2 \\ & \mu_2 \geq -1 \end{aligned}$$

$$\begin{aligned} \mu_1 &\leq 0 \\ \mu_2 &\text{ free} \end{aligned}$$

$$\max 3x_1 + 2x_2 - x_3$$

$$\begin{cases} x_1 - x_2 \geq 2 \\ x_1 + x_3 \geq 5 \end{cases}$$

$$x \geq 0$$

$$\min 2u_1 + 5u_2$$

S.O.

$$u_1 + u_2 \geq 3$$

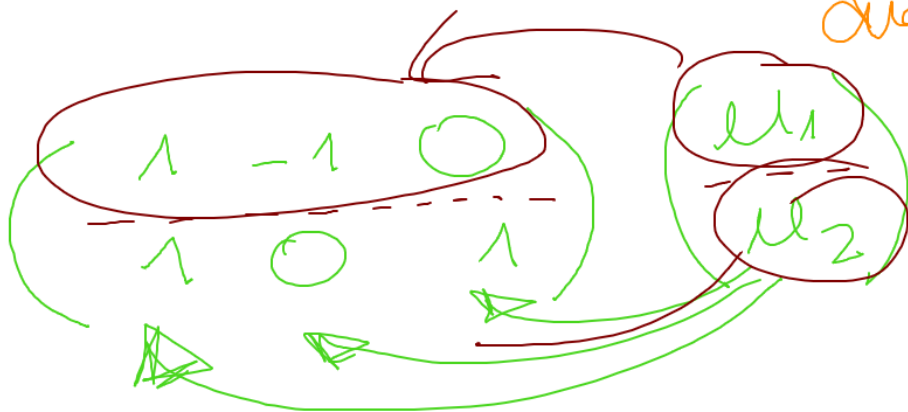
$$-u_1 \geq 2$$

$$u_2 \geq -1$$

$$u_1 \leq 0$$

$$u_2 \text{ free}$$

dual



$$\begin{pmatrix} u_1 & -u_1 & 0 \\ u_2 & 0 & u_2 \end{pmatrix}$$

~~20/02~~

Primal

Restricción $\leq$

Restricción $\geq$

Restricción $=$

Variable ≥ 0

Variable ≤ 0

Variable libre

Dual

Variable ≥ 0

Variable ≤ 0

Variable libre

restriccion ≥ 0

restriccion ≤ 0

restriccion $=$

Ejemplo

max

Cx

$A_1 x \geq b_1$

$A_2 x = b_2$

$x \geq 0$

(u_1)

(u_2)

$\min u_1 b_1 + u_2 b_2$

$u_1 A_1 + u_2 A_2 \geq C$

$u_1 \leq 0$

u_2 libre

$$\begin{bmatrix} \dots & A_1 & \dots \\ \dots & A_2 & \dots \end{bmatrix} \begin{pmatrix} u_1 \\ \dots \\ u_2 \end{pmatrix}$$

Ejms

$$\begin{array}{l} \max Cx \\ Ax \leq b \\ x \geq 0 \end{array} \quad \begin{array}{l} \min yb \\ yA \geq C \\ y \geq 0 \end{array}$$

$$\begin{array}{l} \max Cx \\ Ax = b \\ x \geq 0 \end{array} \quad \begin{array}{l} \min yb \\ yA \geq C \end{array}$$

$$\begin{array}{l} \min Cx \\ Ax \geq b \\ x \geq 0 \end{array} \quad \begin{array}{l} \max yb \\ yA \geq C \\ y \leq 0 \end{array}$$

$$\begin{array}{l} \max Cx \\ Ax \leq b \end{array} \quad \begin{array}{l} \min yb \\ yA = C \\ y \geq 0 \end{array}$$

$$\begin{array}{l} \max Cx \\ Ax = b \end{array} \quad \begin{array}{l} \min yb \\ yA = C \end{array}$$

$$\begin{array}{l} \min Cx \\ Ax \geq b \end{array} \quad \begin{array}{l} \min yb \\ yA = C \\ y \geq 0 \end{array}$$

oo

Volviendo a flujos

$$\max c^T x$$

$$\text{s.t. } Ax = 0 \quad (y)$$

$$I x \leq u \quad (z)$$

$$x \geq 0$$

dual

$$\min z^T u$$

$$\text{s.t. } y^T A + z^T I \geq c$$

$$z \geq 0$$

Dualidad débil:

$$c^T x \leq (y^T A + z^T I) x = y^T (Ax) + z^T (Ix) \leq z^T u$$

$v, w \in \{rs\}$



$$y_w - y_v + z_{vw} \geq 0$$

Analizando el dual

$$\min \sum_e z_e u_e$$



s.u

$$y_w - y_v + z_{vw} \geq 0 \quad \forall vw \in E, v, w \notin \{r, s\}$$

$$\textcircled{0} -y_v + z_{vr} \geq 0 \quad \forall vr, v \notin \{r, s\}$$

$$y_w \textcircled{0} + z_{rw} \geq 0 \quad \forall rw \in E, w \notin \{r, s\}$$

$$-y_v + z_{vs} \geq \textcircled{1} \quad \forall vs, v \notin \{r, s\}$$

$$y_w + z_{sw} \geq \textcircled{-1} \quad \forall sw, w \notin \{r, s\}$$

$$\begin{aligned} z_{rs} &\geq 1 \\ z_{sr} &\geq -1 \\ z_{vw} &\geq 0 \end{aligned}$$

$$\begin{aligned} \hat{y}_s &= 1 \\ \hat{y}_v &= y_v \\ \hat{y}_{\cancel{s}} &= 0 \end{aligned}$$

$$g_r = 1$$

$$g_s = 0$$

$$g_w - g_v + z_{vw} \geq 0$$

$$z_{vw} \geq 0$$